



RAJALAKSHMI INSTITUTE OF TECHNOLOGY
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

ACADEMIC YEAR 2025 - 2026

SEMESTER IV

AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	2117240070006
NAME	S.Abinaya
PROJECT TITLE	Heart Disease Prediction using Logistic Regression
DATE OF SUBMISSION	
FACULTY IN-CHARGE	Mrs. M. Divya

Signature of Faculty In-charge

Signature of HoD

INTRODUCTION

- Machine Learning is a subset of artificial intelligence that enables systems to learn patterns from data and make predictions without explicit programming. In healthcare, machine learning plays a crucial role in predicting diseases and assisting doctors in diagnosis.
- This project focuses on predicting heart disease using Logistic Regression, a supervised learning algorithm used for binary classification. Heart disease is one of the leading causes of death worldwide, and early prediction can help in timely treatment and prevention.
- The aim of this project is to develop a predictive model that analyzes patient medical attributes such as age, cholesterol level, and blood pressure to determine whether a person is likely to have heart disease.

PROBLEM STATEMENT

- To design a machine learning model that predicts the presence or absence of heart disease in a patient based on medical attributes using Logistic Regression.

GOAL

- To develop a predictive model for heart disease detection
- To achieve good accuracy in classification
- To assist in early diagnosis of heart-related problems
- To analyze performance using evaluation metrics

THEORETICAL BACKGROUND

- Machine Learning algorithms are broadly classified into supervised and unsupervised learning. This project uses supervised learning where labeled data is available.
- Logistic Regression is used for binary classification problems. It predicts the probability of an outcome using the sigmoid function.
- **Sigmoid Function:**

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

- A threshold (commonly 0.5) is applied to classify the output:
Probability $\geq 0.5 \rightarrow$ Presence of heart disease
Probability $< 0.5 \rightarrow$ Absence of heart disease
- This approach makes Logistic Regression highly suitable for medical classification problems where outcomes are binary in nature.

ALGORITHM EXPLANATION WITH EXAMPLE

Algorithm

The Logistic Regression algorithm is a supervised learning technique used for binary classification problems. It predicts the probability of a data instance belonging to a particular class (0 or 1) based on input features. It is widely used in medical prediction systems due to its simplicity and interpretability.

Explanation

Logistic Regression works by establishing a relationship between the dependent variable (target) and one or more independent variables (features). It calculates a weighted sum of input features and applies the sigmoid function to convert it into a probability value between 0 and 1. Based on this probability:

- **If the value is greater than or equal to 0.5, the output is classified as 1 (presence of heart disease)**
- **If the value is less than 0.5, the output is classified as 0 (absence of heart disease)**

The model learns these weights during training by analyzing patterns in the dataset.

Mathematical Expression

The probability is computed using the sigmoid function:

$$P(y = 1) = \frac{1}{1+e^{-z}}, \quad z = w_1x_1 + w_2x_2 + \dots + b$$

where:

x = input features
w = weights
b = bias

Example

Consider a patient with the following details:

- **Age = 54**
- **Cholesterol = 240**
- **Resting Blood Pressure = 140**

After applying the trained model:

- **Predicted Probability = 0.76**

Since the probability is greater than 0.5, the model predicts:

1 (Heart Disease Present)

IMPLEMENTATION CODE AND DATASET USED

Dataset Link : <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

Category	Description / Example
Dataset Name	Heart Disease Dataset (Kaggle)
Domain	Healthcare (Medical Diagnosis)
Objective	Binary Classification (Heart Disease Prediction)
Data Source	Public Dataset (Kaggle)
Number of Samples	303 patient records
Data Type	Tabular Data
Instance Description	Each instance represents medical details of one patient
Features (Inputs)	Age, Sex, Chest Pain Type, BP, Cholesterol, Heart Rate, etc.
Feature Types	Numerical and Categorical text
Target Variable	Presence of Heart Disease (0 = No, 1 = Yes)
Number of Classes	2 Classes (Binary)
License Version	Open Public Dataset
Version	Version 1 (Standard Kaggle Dataset)

Python Code :

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load dataset
data = pd.read_csv("heart.csv")

# Features and target
X = data.drop("target", axis=1)
y = data["target"]

```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Model
model = LogisticRegression(max_iter=1000)

# Train
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Results
print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Plot confusion matrix
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt="d")
plt.show()
```

OUTPUT

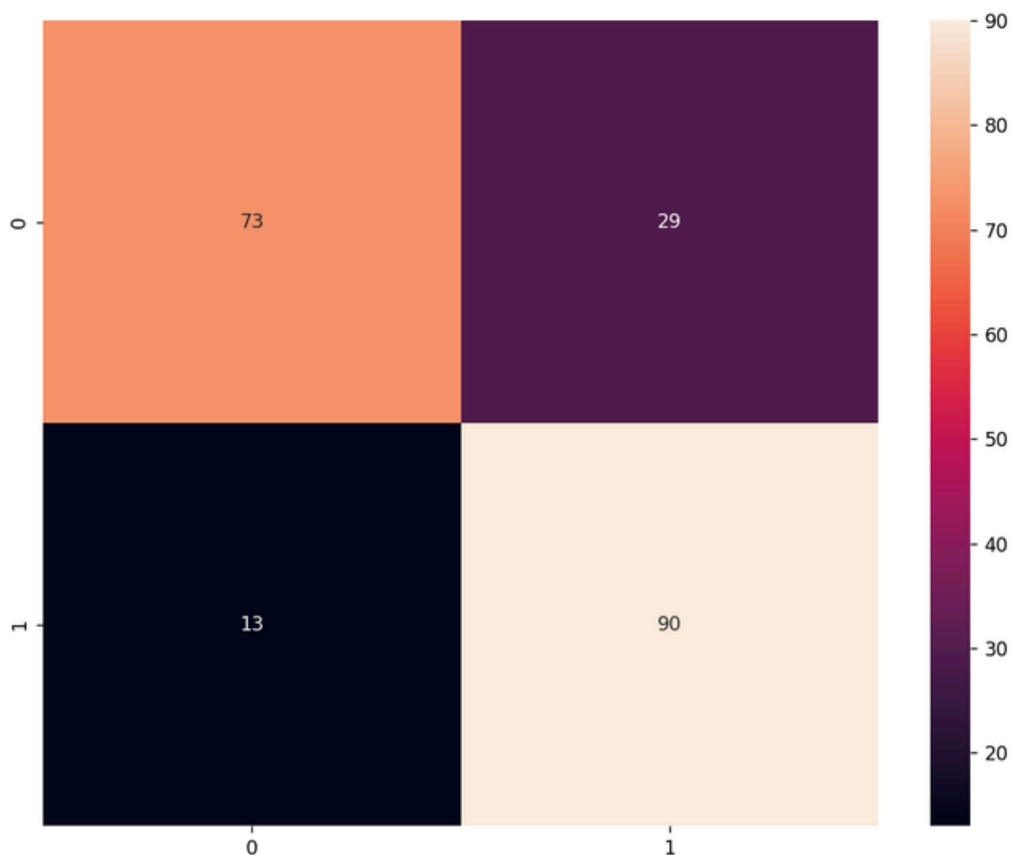
Accuracy: 0.7951219512195122

Confusion Matrix:

```
[[73 29]
 [13 90]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.72	0.78	102
1	0.76	0.87	0.81	103
accuracy			0.80	205
macro avg	0.80	0.79	0.79	205
weighted avg	0.80	0.80	0.79	205



RESULTS AND FUTURE ENHANCEMENT

Results

- The Logistic Regression model was successfully implemented to predict the presence of heart disease using patient medical data. The model achieved an accuracy of approximately 80%, indicating good predictive performance for a basic classification approach.
- The confusion matrix shows that most of the instances were correctly classified, with a higher number of true positives and true negatives. The classification report also indicates balanced precision and recall values, suggesting that the model performs consistently across both classes.
- Overall, the model demonstrates that Logistic Regression can effectively handle healthcare-related binary classification problems with satisfactory results.

Future Enhancements

- Improve model accuracy by using advanced algorithms such as Random Forest or Support Vector Machine
- Apply feature selection and data preprocessing techniques to enhance performance
- Use larger and more diverse datasets for better generalization
- Develop a user-friendly interface for real-time prediction
- Deploy the model as a web or mobile application for practical use
- Integrate additional medical parameters to improve prediction accuracy

Git Hub Link of the project	https://github.com/abinaya22086/ML-Mini-Project
------------------------------------	---

REFERENCES

- 1., Christopher M. Bishop, Springer, 2006.
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Aurélien Géron, O'Reilly Media, 2019.
3. Scikit-learn Documentation – <https://scikit-learn.org/stable/>
4. Heart Disease Dataset (Kaggle) – <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>
5. GeeksforGeeks – Machine Learning Tutorials, <https://www.geeksforgeeks.org/machine-learning/>